SESSION 4 — AFTERNOON

Equity Investments

Market structure • Equity securities • Industry analysis • Equity valuation

Session Roadmap — 2 Hours

Equity investments are the most complex and rewarding part of the CFA curriculum. This session covers markets, securities and valuation — all from first principles.

01 **Market Organisation**
Market types • order types • market indices • index construction

02 **Market Efficiency**
Weak/semi-strong/strong forms • EMH implications • anomalies • behavioural biases

03 **Equity Securities**
Common vs preferred shares • ADRs • ROE and equity value linkage

04 **Industry & Competitive Analysis**
Industry classification • Porter's Five Forces • competitive positioning

05 **Equity Valuation: DDM**
Gordon Growth: D0=€5 g=4% r=8% → V=€130 (CFA LM8) • preferred stock: $4.75/7.5%=$63.33

06 **Equity Valuation: Multiples**
P/E (justified, Canon trend), P/B, P/S, EV/EBITDA • Telefónica vs Deutsche Telekom

Learning Objectives

- **Market Organisation & Index Types**
  Describe exchange-traded vs OTC markets. Classify order types. Distinguish price-weighted, market-cap-weighted and equal-weighted indices.
- **Market Efficiency**
  Define and apply weak, semi-strong and strong form EMH. Explain implications for technical/fundamental analysis and active management.
- **Equity Security Characteristics**
  Distinguish common from preferred shares. Explain ADRs/GDRs. Connect ROE to book value growth and intrinsic value.
- **Industry & Competitive Analysis**
  Apply Porter's Five Forces. Assess industry lifecycle. Identify competitive advantages and moats.
- **Dividend Discount Models**
  Calculate Gordon Growth Model value (D0=€5, g=4%, r=8% → €130). Value preferred stock (Union Electric → $63.33). Apply two-stage DDM.
- **Price Multiples & EV/EBITDA**
  Compute and interpret P/E, P/B, P/S, P/CF and EV/EBITDA. Apply CFA Canon and Telefónica/Deutsche Telekom examples.

SECTION 01

Market Organisation (CFA LM1)

Market Organisation — Types and Functions (CFA LM1)

Financial markets connect buyers and sellers of financial assets. CFA LM1: markets serve three core functions — allocate savings, discover prices, enable risk transfer.

Market Type	Definition	Example	Key Feature
Exchange-traded market	Centralised venue with standardised contracts	NYSE, LSE, CME (futures)	Transparent prices, regulated, counterpa via clearing
Over-the-counter (OTC)	Bilateral trades between dealers, no central venue	FX market, bond markets, OTC derivatives	Customisable, less transparent, counterpa
Primary market	New securities issued to public for first time	IPO, SEO, bond offering	Proceeds go to issuer (capital raising)
Secondary market	Existing securities traded between investors	Stock exchange trading, OTC bond trading	Provides liquidity; proceeds go to seller, r issuer
Quote-driven (dealer) market	Dealers post bid/ask prices; investors trade against them	OTC bond market, FX interbank	Dealers provide liquidity; earn bid-ask spr
Order-driven market	Orders from all participants matched by exchange rules	Most equity exchanges	Transparent order book; competitive pric

Order Types — Execution, Validity and Size (CFA LM1)

Every order specifies: WHAT to buy/sell, HOW MUCH, at WHAT PRICE, for HOW LONG

Order Type	Execution Rule	Guaranteed Fill?	Price Certainty?	Use When
Market order	Execute immediately at best available price	Yes (if liquidity exists)	No: price depends on market	Speed > price certainty. Urgent.
Limit order	Execute only at specified price or better	No: may not fill	Yes: set maximum (buy) or minimum (sell)	Price matters. Patient investor.
Stop (stop-loss) order	Becomes market order when stop price is hit	No: triggered, then market	No: converts to market order	Protect gains or limit losses
Stop-limit order	Becomes limit order when stop price is hit	No	Partial: limits how far price goes	More control but risk of no fill
Market-on-close	Execute at closing auction price	Yes (at close)	Near-closing price	Index funds matching benchmark
Good-till-cancelled (GTC)	Stays active until filled or cancelled	If price is reached	Yes (limit)	Long-term tactical orders

Security Market Indices — Construction Methods (CFA LM2)

A market index tracks the performance of a defined set of securities. The construction method determines what the index actually measures.

Weighting Method	Formula	Advantage	Limitation	Example
Price-weighted	Simple average of member prices	Easy to calculate	Higher-price stocks dominate. Stock splits distort.	Dow Jones Indu Average
Market-cap-weighted	Weight = market cap / total market cap	Reflects economic importance	Largest companies dominate; momentum bias	S&P 500, FTSE 1 MSCI World
Float-adjusted market cap	Use only freely tradable shares, not insider holdings	More realistic tradable exposure	Excludes government/founding family stakes	Most major ind today
Equal-weighted	Same weight to every constituent	Captures small-cap exposure	Requires frequent rebalancing; impractical at scale	SPDR Equal We 500
Factor-weighted	Weight by fundamental factors (value, quality, etc.)	Targets specific risk premiums	Potentially higher turnover, costs	Smart beta / fa

Short Selling & Leveraged Positions (CFA LM1)

Short position: borrow shares → sell → buy back later. Profit if price FALLS. Loss unlimited if price rises.

CFA LM1: investors can take leveraged (margin) positions or short positions. Both amplify returns but also amplify risk.

Short Selling — Mechanism

1. Borrow shares from broker (pay borrow fee).
2. Sell immediately in market at current price.
3. Wait for price to fall.
4. Buy back at lower price.
5. Return shares to lender. Profit = Sale price − Repurchase price − Costs.

Risk: unlimited loss (price can rise indefinitely). Short squeeze: forced buying by many shorts simultaneously.

Leveraged (Margin) Positions

Borrow funds from broker to buy MORE shares than cash allows.
Initial margin: minimum equity % required (e.g. 50%).
Maintenance margin: minimum ongoing (e.g. 25%).
Margin call: if equity < maintenance, must deposit more or sell.

Leverage amplifies gains: 2x leverage on +10% move = +20%.
Leverage amplifies losses: 2x leverage on −10% = −20%.

CFA LM1: leverage ratio = value of position / equity invested. A margin loan of $40,000 on $100,000 position = 2.5x leverage. Total loss possible exceeds initial investment if leverage used.

SECTION 02

Market Efficiency (CFA LM3)

Efficient Market Hypothesis — Three Forms (CFA LM3)

EMH: prices fully and immediately reflect all available information. Active management cannot consistently earn excess risk-adjusted returns.

Form	Information Reflected in Prices	Implication for Analysis	Evidence
Weak form	All PAST trading prices and volume (historical)	Technical analysis does NOT work (patterns are random). Fundamental analysis may work.	Mixed: some short-term momentum anomalies found
Semi-strong form	All PUBLICLY available information (prices, reports, news, analyst forecasts)	Technical AND fundamental analysis do not work consistently. Only insider info would help.	Generally supported: markets adjust quickly to public news. Most active managers underperform.
Strong form	ALL information: public AND private (insider) information	Even insiders cannot earn excess returns consistently. Perfection.	Rejected: insider trading cases prove private info creates edge (but illegal)

Implications for Active Managers

Semi-strong EMH implies active management (stock picking, fundamental research) cannot consistently beat the market after fees.

Response: either markets are not semi-strong, or there are persistent mispricings (anomalies, illiquid/neglected securities)

EMH Caveats & Anomalies

Value premium: low P/B stocks historically outperformed.

Size premium: small-cap outperformed.

Momentum: past 12-month winners beat losers.

These may be: (a) risk premia, (b) data mining, or (c) genuine inefficiency

Behavioural Finance — Why Markets May Be Inefficient (CFA LM3)

CFA LM3: behavioural finance challenges EMH by showing investors systematically deviate from rationality in predictable ways.

Bias	Description	Investment Impact	Example
Loss aversion	Pain of losses felt 2× more than pleasure of equivalent gains	Hold losers too long (avoid realising loss). Sell winners too early.	Disposition effect: investor won't sell a stock down
Overconfidence	Investors overestimate their own skill and knowledge accuracy	Over-trade, under-diversify, concentrated bets	Active traders typically underperform passive after
Herding	Investors mimic crowd behaviour rather than independent analysis	Bubbles (all buy) and panics (all sell)	Tech bubble 1999–2000, Meme stocks 2021
Anchoring	Over-reliance on first piece of information (anchor) in decision making	Stick to initial price paid, ignoring new information	Won't sell below purchase price even if fundament collapse
Representativeness	Judge probability based on similarity to known patterns	Chase recent performance; confuse good company with good stock	Buy fund after great 3-year run (always lagged)
Confirmation bias	Seek information that confirms existing view, ignore contradictory	Insufficient due diligence; underweighted red flags	Analyst bullish on stock ignores bearish analyst rep

Active vs Passive Investment — EMH Implications (CFA LM3)

CFA LM3: if markets are semi-strong efficient, active management (stock picking) cannot consistently outperform after fees. This debate shapes the entire investment industry.

Approach	Premise	Typical Cost	Evidence
Passive (index) investing	Market is semi-strong efficient. Best return = market return minus fees.	0.03–0.20% TER	SPIVA data: 85%+ of active managers underperform over 10 years
Active fundamental	Semi-strong EMH wrong. Research uncovers mispricings.	0.5–1.5% TER + transaction costs	Some managers outperform but not persistently. Hard to distinguish skill from luck.
Active quantitative	Systematic factors can be exploited (value, momentum, quality)	0.2–0.8% TER	Factor premia documented academically. But may be eroded once widely known.
Private markets	Inefficient; illiquid; superior due diligence adds value	1–2% management + 20% carry	Illiquidity premium real but difficult to access; long lock-ups

The Paradox of Active Management

Markets BECOME efficient because active managers trade on mispricings and correct them. If everyone went passive, no-one would correct mispricings and markets would become inefficient.

Gross-of-fees: the average active manager must earn the index return (zero-sum game minus transactions).

Net-of-fees: the average active manager MUST underperform passive after fees. The question is whether the BEST managers consistently outperform by enough to justify their fees.

SECTION 03

Equity Securities (CFA LM4)

Common vs Preferred Equity — CFA LM4

Common = ordinary shares (residual). Preferred = priority claim, fixed dividend, usually no vote.

Feature	Common Shares	Preferred Shares
Dividend	Variable: paid from residual profits, at board's discretion	Fixed: stated rate on par value. Cumulative if missed.
Priority	Last in line: receives what's left after all other claims	Before common, after debt. Priority in dividends and liquidation.
Voting rights	Vote on board elections, major corporate transactions	Usually non-voting (in exchange for priority)
Capital appreciation	Full upside: profits from growing earnings	Limited: priced like bond. Less upside but more stable.
Return characteristics	Equity risk premium. Higher expected return.	Hybrid: equity-like structure, bond-like return profile.
Typical investors	Equity investors: pension funds, mutual funds, individuals	Income investors, insurance companies, preferred ETFs

CFA LM4: preferred shares may also be CONVERTIBLE (into common shares), CALLABLE (issuer can redeem), CUMULATIVE (unpaid dividends accumulate) or PARTICIPATING (share in excess earnings).

ROE and Equity Value — Why ROE Drives Intrinsic Value (CFA LM4)

CFA LM4: equity investors receive the benefit of a company's future earnings. ROE links accounting performance to the intrinsic value of equity.

Book Value per Share$_t$ = Book Value$_{t-1}$ + EPS − DPS (grows by retained earnings)

ROE and the SUSTAINABLE GROWTH RATE

g (sustainable growth rate) = b × ROE
where b = retention rate = 1 − payout ratio

Example: ROE = 15%, payout ratio = 40%, retention rate b = 60%.
g = 0.60 × 15% = 9%

Intuition: the company retains 60% of earnings and earns 15% on that equity. So equity book value grows at 9% per year. If dividends also grow at 9%, this g can be used in the Gordon Growth Model.

CFA insight: a company can increase g by (1) improving ROE (better profitability/efficiency/leverage) or (2) retaining more earnings (reducing payout ratio). But reducing payout also reduces dividend, so these trade off in the DDM.

Cost of Equity vs ROE

If ROE > r_e (cost of equity): company earns above required return. Creates value. P/B > 1.0.
If ROE = r_e: earns exactly required return. P/B = 1.0.
If ROE < r_e: destroys value. P/B < 1.0

ADRs and GDRs

American/Global Depositary Receipts: foreign shares held by bank, USD/denominated certificates issued to US investors. Eliminate FX settlement. Allow exposure to foreign companies on home exchange.

Equity Return & Risk Characteristics (CFA LM4)

CFA LM4: equity securities are the most volatile but historically highest-returning asset class. Understanding return components and risks is fundamental.

Total Return = Price Appreciation + Dividend Yield (plus FX if non-domestic)

Component	Description	Range (Long Run)	Influenced By
Dividend yield	Annual dividend / current price	1–5% for dividend payers; 0% for growth stocks	Payout policy, earnings stability, divide growth expectations
Price appreciation	Capital gain from rising market price	4–8% historically in developed markets	EPS growth, P/E re-rating, macro cond
Total equity return	Historical equity risk premium vs bonds	5–8% above risk-free rate (long-run average)	GDP growth, inflation, risk aversion, va
Currency effect	FX change on foreign holdings	+/−10%+ per year	Relative inflation, interest rates, trade

Equity Risks vs Other Asset Classes

Systematic risk (β): cannot be diversified away. Compensated by equity risk premium.

Unsystematic (specific) risk: can be eliminated via diversification across 20+ uncorrelated securities.

Equity subordination: in liquidation, equityholders are LAST. In distress, equity often worth zero.

Liquidity risk: small/mid-cap stocks can be hard to sell in quantity without moving price.

SECTION 04

Industry & Competitive Analysis (CFA LM6)

Industry Analysis — Why It Matters for Equity Analysts (CFA LM6)

CFA LM6: industry characteristics dominate individual company performance. Companies in attractive industries have structural tailwinds; those in challenged industries fight headwinds.

- **Improve earnings forecasts**
  Industry economics (pricing, cost structure, competitive intensity) determine sustainable margins. Analyse industry BEFORE company.
- **Identify investment opportunities**
  Top-down: find attractive industries first, then pick the best company within them. Sector rotation is a key active strategy.
- **Assess competitive position**
  Same company can thrive or fail depending on industry structure. Industry is the sea; company positioning is the boat.
- **Benchmark company performance**
  Peer comparison only meaningful within the same industry. Compare margins, growth and capital allocation to industry average.
- **Understand macro-industry linkages**
  Different industries respond differently to business cycles. Cyclical vs defensive vs growth vs interest-rate-sensitive.

Porter's Five Forces — Industry Profitability Framework (CFA LM6)

CFA LM6: Michael Porter's Five Forces determine the long-run profitability potential of an industry. All five must be assessed together.

1. Threat of New Entrants

LOW → high barriers = high profit potential

Barriers: capital requirements, economies of scale, brand loyalty, patents, switching costs, regulatory licences. Example: commercial aircraft (HIGH barrier, Airbus/Boeing duopoly). Coffee shops (LOW barrier).

2. Bargaining Power of Suppliers

LOW → many suppliers = high profit potential

Strong if: few suppliers, differentiated inputs, high switching costs, suppliers can forward-integrate. Example: TSMC has high power over chip customers. Wheat farmers have low power.

3. Bargaining Power of Buyers

LOW → fragmented buyers = high profit potential

Strong if: few large buyers, standardised product, low switching cost, buyers can backward-integrate. Example: Walmart has huge power over suppliers. Individual consumers have low power.

4. Threat of Substitutes

LOW → few substitutes = high profit potential

Substitutes reduce pricing power. Electric vehicles substitute for petrol cars. Video calls substitute for business travel. The more substitutes, the lower the sustainable price premium.

5. Competitive Rivalry

LOW rivalry = high profit potential

Intense if: many competitors, slow growth, high fixed costs, undifferentiated products, high exit barriers. Steel, airlines, retail banking = high rivalry. Pharmaceuticals (patented drugs) = low rivalry.

Industry Lifecycle — Growth, Margins & Strategy (CFA LM6)

Industries evolve through predictable stages. Understanding the lifecycle stage helps forecast revenue growth, margin trajectory and competitive dynamics.

Stage	Revenue Growth	Profitability	Competition	Investment Approach
Embryonic (launch)	Very high (small base)	Negative/break-even	Few, innovative firms	VC: high risk, no earnings. Binary outc
Growth	Rapid (20%+)	Improving, reinvested	New entrants, market expands	P/S, EV/Sales: no earnings yet. Pay for optionality.
Shakeout	Decelerating	Competitive pressure	Consolidation. Weaker firms exit.	Select quality: survivors take share.
Mature	GDP-like (low single digits)	Stable, returns capital	Established oligopoly	P/E, dividends. EV/EBITDA multiples fa
Decline	Negative or flat	Contracting	Shrinkage. Survivors	Asset value, FCF yield. Value trap risk.

Investment implications by stage

Growth stage companies: justify high multiples if growth rate is high enough. Pay for future earnings, not current.
Mature stage companies: value based on dividends and P/E. EV/EBITDA vs peers.
Declining stage: watch for disruption. Book value and asset liquidation relevant. Dividend sustainability key risk.

Competitive Positioning — Moats & Pricing Power (CFA LM6)

CFA LM6: beyond industry structure, individual company competitive positioning determines ability to earn above-normal returns within an industry.

Competitive Advantage	Description	Examples	Durability
Cost leadership	Lowest cost producer. Pricing power = can undercut and still profit.	Amazon logistics, TSMC fabs, Ryanair	High if structural. Eroded by innovation.
Differentiation / Brand	Premium pricing via unique product or brand loyalty.	Apple iPhone, LVMH, Hermès, Nike	High if brand is deeply embedded. Can erode.
Network effects	Product becomes more valuable as more users join. Winner-takes-most.	Visa/Mastercard, LinkedIn, WhatsApp	Very high once network established.
Switching costs	Customers face high cost to change suppliers.	SAP ERP, Bloomberg Terminal, Salesforce CRM	High. Customers sticky even with better alternatives.
Intangibles / IP	Patents, regulatory licences, proprietary data or processes.	Pharma patents, regulatory exclusivity, data assets	Medium-high. Patent cliff is a risk.
Scale advantages	Fixed costs spread over larger volume = lower unit cost.	Walmart, Google Search, Netflix	High. Hard to replicate without matching scale.

Morningstar Moat Ratings: Wide moat = competitive advantage likely to persist 20+ years. Narrow moat = 10+ years. No moat = unlikely to last. Moat strength directly predicts ability to sustain ROIC > WACC.

SECTION 05

Equity Valuation: DDM (CFA LM8)

Equity Valuation Framework — Three Approaches (CFA LM8)

CFA LM8: analysts estimate INTRINSIC VALUE and compare to market price. Overvalued if V < P; undervalued if V > P.

Approach	Core Logic	Models	Best For	Limitation
Present Value (Intrinsic)	Discount future cash flows to PV	DDM, FCFE, FCFF	Dividend-paying, stable firms	Sensitive to growth assumptions
Price Multiples	Compare price to a value driver using peers	P/E, P/B, P/S, EV/EBITDA	Screening, relative comparison	Requires comparabl comparable
Asset-Based	Value underlying assets, net of liabilities	Net asset value, book value, liquidation value	Banks, real estate, holding companies, distressed	Historical cost may u current value

Why no single model is enough

CFA LM8: 'Analysts must cope with uncertainties related to model appropriateness and the correct value of inputs.'
Best practice: triangulate across 2-3 models. If all methods point to undervaluation, conviction is higher.
Professional investors use DCF (anchor), EV multiples (quick benchmark), and NAV/asset-based (floor value in stress).

Preferred Stock Valuation — CFA LM8 (Union Electric Example)

$V_0 = D_0 / r$ (perpetuity formula for non-callable, non-convertible preferred stock)

Preferred stock pays a fixed dividend indefinitely (if perpetual). It is valued exactly like a perpetuity in TVM notation.

CFA LM8 EXAMPLE 6 — UNION ELECTRIC CO. 4.75% PERPETUAL PREFERRED

Data: Par value = USD 100. Annual dividend = USD 4.75 (4.75% × $100). Maturity = perpetual. No call, no conversion.
Required rate of return on Ba1/BB-rated preferred shares = 7.5%.

Intrinsic value = D/r = USD 4.75 / 0.075 = USD 63.33

Since market price < $100 par value, the issue is trading at a large DISCOUNT to par.
Callability: if the issue were callable, the issuer would only call when market price > call price. Since price ($63.33) << par ($100), callability is irrelevant here and barely reduces value.

CFA LM8 Case 1 answer: USD 63.33. CFA LM8 adds: callable issues have lower intrinsic value (call option beneficial to issuer, not investor).

Gordon (Constant) Growth Model — CFA LM8

$$V_0 = D_1 / (r - g) = D_0(1+g) / (r - g) \quad [r = \text{required return}, g = \text{constant growth rate}]$$

The Gordon Growth Model values a stock as a growing perpetuity of dividends. Requires: stable, perpetual business; r > g always.

CFA LM8 — CORE EXAMPLE

Data: Current annual dividend D_0 = €5.00. Constant growth rate g = 4%. Required return r = 8%.

$D_1 = D_0 \times (1+g)$ = €5.00 × 1.04 = €5.20
$V_0 = D_1 / (r - g)$ = €5.20 / (0.08 − 0.04) = €5.20 / 0.04 = €130 per share

CRITICAL NOTE: Numerator is D_1 (next year's dividend), NOT D_0. CFA: 'Using the wrong numerator is a common error.'

Sensitivity: if g rises from 4% to 5%: V_0 = 5.25/0.03 = €175 (+35%). If r rises from 8% to 9%: V_0 = 5.20/0.05 = €104 (−20%). Growth rate changes are highly amplifying!

Appropriate when: company in mature, stable stage. Utility companies, consumer staples, established financial institutions. NOT for high-growth tech.

Sustainable Growth Rate & Two-Stage DDM (CFA LM8)

$$g = b \times ROE \quad \text{where } b = \text{retention rate} = 1 - \text{dividend payout ratio}$$

Most companies cannot maintain a single constant growth rate forever. The two-stage DDM handles a high-growth phase followed by mature steady-state growth.

TWO-STAGE DDM WORKED EXAMPLE

Company: ROE = 18%, payout = 40%, b = 60%. Current dividend D_0 = \$2.00. Required return r = 12%.
High-growth phase $g_1 = b \times ROE = 60\% \times 18\% = 10.8\%$ for 5 years.
Mature phase g_2 = 4% thereafter.

Year 1: $D_1 = 2.00 \times 1.108 = \$2.216$ Year 2: $D_2 = 2.216 \times 1.108 = \$2.455$
Year 3: \$2.720 Year 4: \$3.014 Year 5: \$3.339
Terminal value at end of Year 5: $TV_5 = D_6/(r-g_2) = (3.339\times1.04)/(0.12-0.04) = 3.473/0.08 = \$43.41$

V_0 = PV of years 1–5 + PV(TV_5) at 12%
$= 2.216/1.12 + 2.455/1.12^2 + 2.720/1.12^3 + 3.014/1.12^4 + (3.339+43.41)/1.12^5$
$= 1.979 + 1.958 + 1.935 + 1.916 + 26.50 = \$34.29$

SECTION 06

Equity Valuation: Price Multiples (CFA LM8)

Price-to-Earnings Ratio — The Most Cited Multiple (CFA LM8)

P/E = Price per Share / EPS Justified P/E = p / (r − g) = payout ratio / (r − g)

P/E is the most widely cited valuation multiple. CFA LM8: P/E is positively related to growth and payout ratio, inversely to required return.

Justified P/E from Gordon Growth Model

From Gordon Growth: $P_0 = D_1/(r-g)$. Divide both sides by E_1 (forward EPS): $P_0/E_1 = (D_1/E_1)/(r-g) = p/(r-g)$
where p = payout ratio = D_1/E_1.

Example: payout ratio p = 40%, r = 10%, g = 5%.
Justified forward P/E = 0.40 / (0.10 − 0.05) = 0.40 / 0.05 = 8.0x

If payout rises to 60%: P/E = 0.60/0.05 = 12.0x (more generous dividend → higher P/E)
If r rises to 12%: P/E = 0.40/0.07 = 5.7x (higher discount rate → lower P/E)
If g rises to 7%: P/E = 0.40/0.03 = 13.3x (faster growth → much higher P/E)

CFA caution: P/E is meaningless when EPS is negative. Use P/S or EV/EBITDA instead.

P/E Trend Analysis — CFA LM8 Canon Inc. (Example 15)

Directly from CFA Vol 5, LM8, Example 15. Canon P/E trend analysis over 5 years reveals valuation context.

Year	Price (JPY)	EPS (JPY)	P/E	Analyst Signal
2013	¥3,330	¥200.8	16.6x	Historical low — floor of 5-year range
2014	¥3,840.5	¥229.0	16.8x	Consistent with 2013 level
2015	¥3,675	¥201.7	18.2x	Slight re-rating upward
2016	¥3,295	¥138.0	23.9x	PEAK — EPS collapsed, price held. P/E inflated.
2017	¥4,200	¥222.9	18.8x	Near historical average. EPS recovered. Fairly pric

CFA LM8 Analysis

Trend analysis: 2016 peak P/E of 23.9x reflects EPS collapse, not re-rating. Price held while earnings fell.

At year-end 2017 P/E of 18.8x: close to historical average of 16.6–18.2x. Appears fairly priced based on trend.

Bull case: P/E re-rating above 18.8x justified by EPS recovery. Bear case: P/E reverts to historical low of 16.6x → price = 16.6 × ¥222.9 = ¥3,700. Downside limited if EPS holds.

Price Multiples Suite — P/B, P/S, P/CF and EV/EBITDA (CFA LM8)

Different multiples highlight different value drivers. CFA LM8 tests all four. Use multiple together for a fuller picture.

Multiple	Formula	Best For	Limitation
P/E (Price/Earnings)	Stock price / EPS	Profitable companies	Meaningless if EPS negative. Affected by leverage.
P/B (Price/Book)	Stock price / Book value per share	Banks, financials, asset-heavy	Book value may not reflect market value of assets. Intangibles excluded.
P/S (Price/Sales)	Stock price / Revenue per share	Loss-making companies, revenue visibility	No profitability info. High revenue ≠ high earnings.
P/CF (Price/Cash Flow)	Stock price / Operating cash flow per share	When earnings distorted by D&A	Cash flow can still be manipulated via working capital.
EV/EBITDA	Enterprise value / EBITDA	Cross-border, cross-capital-structure comparison	EBITDA ignores capex. High D&A companies look cheaper.

CFA LM8: Telefónica vs Deutsche Telekom — Multi-Multiple Comparison

CFA Vol 5, LM8. Two European telecoms. Multiples analysis reveals mixed signals. Requires integration rather than reliance on a single metric.

Multiple (2017)	Telefónica	Deutsche Telekom	Interpretation
P/Revenue (P/S)	0.8x	0.8x	Equal. Both in line with revenue.
P/Cash Flow (P/CF)	2.9x	3.7x	Telefónica cheaper on cash flow. Better relative val
P/Book (P/B)	2.4x	1.5x	Deutsche Telekom cheaper on book. More assets p
P/Earnings (P/E)	14.0x	18.2x	Telefónica cheaper on earnings. But 2016 P/E was 2
EV/EBITDA	2.5x	3.1x	Telefónica cheaper by EV/EBITDA. This EBITDA diffe D&A differences — DT appears to have higher earn

CFA LM8 Conclusion

Multiples broadly suggest Telefónica is attractively valued vs Deutsche Telekom on P/CF, P/E and EV/EBITDA. BUT: P/B suggests the opposite (DT cheaper). The analyst must investigate WHY — difference in asset base quality? Accounting differences? Higher DT leverage?

Key insight: the P/E gap (14x vs 18x) cannot be explained by higher DT depreciation charges, higher interest, or taxes. Appears to

Enterprise Value — Capital-Structure-Neutral Valuation (CFA LM8)

EV = Market Cap + Preferred + Total Debt − Cash & Equivalents

EV = the cost of a complete takeover: you assume the target's debt but receive its cash. EV is capital-structure-neutral — it removes the effect of leverage.

Why EV is Used in M&A

Acquiring company inherits ALL capital claims.
EV captures the full acquisition cost: pay for equity, assume debt, receive cash.

Example: Target market cap = £300m. Debt = £200m. Cash = £50m.
EV = 300 + 200 − 50 = £450m (true cost to acquirer).

Without EV, a company with huge debt would look cheap on P/E but is actually expensive.

EV/EBITDA in Practice

EBITDA = proxy for pre-D&A, pre-interest operating cash flow.
EV/EBITDA compares enterprise value to operating cash.

Advantage: not affected by D&A method, leverage or tax rate → cross-border and cross-capital-structure comparable.

Typical ranges: Utilities 8-12x. Consumer staples 12-18x. Tech 20-40x+.
CFA: 'EV/EBITDA is arguably the most common EV multiple in Europe.'

EV multiples most useful when P/E is distorted by: (a) different D&A policies, (b) different tax rates, (c) different capital structures. Always check Net Debt / EBITDA alongside EV/EBITDA.

Asset-Based Valuation — When to Use NAV (CFA LM8)

NAV = Fair Value of Assets − Fair Value of Liabilities (net asset value per share = NAV / shares)

CFA LM8: asset-based models estimate value by marking assets to fair value. Particularly relevant for holding companies, financials and real estate.

Situation	Why Asset-Based Preferred	Example	Note
Investment holding companies	Value = sum of portfolio holdings at market prices	Berkshire Hathaway: sum of listed + private stakes	Conglomerate discount: market values less than NAV
Real estate / REITs	Properties can be valued by independent appraisers	British Land, Unibail. NAV = property values − debt	P/NAV premium or discount vs peers is key signal
Banks / insurers	Financial assets mainly: loans, securities, derivatives	P/B is proxy for asset-based. Marks matter.	Loan quality (NPLs) and mark-to-model risk critical
Distressed companies	Liquidation value as floor, not going-concern value	Chapter 7 liquidation analysis	Enterprise value may be below book. Equity = zero if liabilities > assets.
Natural resources	Reserves valued independently; asset value key	Oil company: PV10 of proved reserves × oil price	Reserve life index (RLI) = reserves/production

CFA LM8: asset-based valuation is most reliable when assets are liquid, market values are observable and the business model primarily holds assets rather than creates them.

PRACTICE PROBLEMS

Work through each before checking answers.

Q1 Gordon Growth Model (CFA LM8): D_0 = €3.00, g = 5%, r = 9%. (a) Calculate D_1. (b) Calculate intrinsic value V_0. (c) If market price = €82, is the stock overvalued, undervalued or fairly valued?

Q2 Preferred stock (CFA LM8, Union Electric): A non-callable perpetual preferred stock pays USD 4.75/year. Required return = 7.5%. What is the intrinsic value?

Q3 Porter's Five Forces: A pharmaceutical company has: (a) patent protection on its blockbuster drug; (b) no close substitutes; (c) buyers are governments with price negotiation power; (d) active patent expiry (generic entry). Assess which forces are strong vs weak.

Q4 Justified P/E (CFA LM8): A company has payout ratio = 50%, required return r = 11%, sustainable growth rate g = 5%. Calculate the justified forward P/E. If current forward P/E = 14x, is the stock over/undervalued?

Q5 EV calculation: Company A: market cap = $500m, total debt = $300m, cash = $80m, EBITDA = $120m. Company B: market cap = $400m, no debt, cash = $50m, EBITDA = $80m. Calculate EV and EV/EBITDA for each. Which is cheaper on EV/EBITDA?

Practice — Worked Solutions

A1

(a) $D_1 = D_0 \times (1+g) = 3.00 \times 1.05 = €3.15$. (b) $V_0 = D_1/(r-g) = 3.15/(0.09-0.05) = 3.15/0.04 = €78.75$. (c) Market price = €82 > intrinsic value €78.75. Stock is slightly OVERVALUED (by 4%). Analyst would recommend hold or sell depending on conviction level in model inputs.

A2

$V_0 = D/r$ = USD 4.75 / 0.075 = USD 63.33 (CFA LM8 Example 6 Case 1). This is the present value of a perpetuity of $4.75/year discounted at 7.5%.

A3

Five Forces for pharma: (1) New entrants: WEAK (high barriers: patents, clinical trials, FDA approval). (2) Supplier power: WEAK (pharma firms are large buyers of API). (3) Buyer power: STRONG (government payers negotiate). (4) Substitutes: WEAK while patent exists, then STRONG when generic entry occurs. (5) Rivalry: WEAK (patent protection). Overall: HIGH profitability while patented. Cliff risk at patent expiry.

A4

Justified P/E = $p/(r-g) = 0.50/(0.11-0.05) = 0.50/0.06 = 8.33$x. Current P/E = 14x > justified 8.33x. Stock appears OVERVALUED based on these fundamentals. Market may be pricing in higher growth, lower required return, or higher payout than assumed.

A5

Company A: EV = 500+300−80 = $720m. EV/EBITDA = 720/120 = 6.0x. Company B: EV = 400+0−50 = $350m. EV/EBITDA = 350/80 = 4.4x. Company B is CHEAPER on EV/EBITDA (4.4x vs 6.0x) despite paying higher price-to-market-cap ratio. This shows why EV matters: Company A's debt load makes it more expensive in true acquisition cost terms.

Formula Reference Card — Session 8 Equity

Key equity valuation formulas from this session.

Perpetual preferred stock	$V_0 = D / r$
Gordon Growth Model	$V_0 = D_1 / (r-g) = D_0(1+g) / (r-g)$ [D_1 NOT D_0 in numerator]
Sustainable growth rate	$g = b \times ROE$ where b = retention rate = 1 − payout ratio
Justified forward P/E	$P_0/E_1 = p / (r-g)$ where p = payout ratio
CAPM (required return)	$r = r_f + \beta \times (r_m - r_f)$
Enterprise Value	EV = Market Cap + Preferred + Total Debt − Cash
EV/EBITDA	EV / EBITDA (capital-structure neutral; cross-country comparable)
P/E (trailing)	Current price / Last 12 months EPS
P/E (forward)	Current price / Next 12 months forecast EPS
P/B	Price per share / Book value per share
P/S	Price per share / Revenue per share
Two-stage DDM	PV(dividends in high-growth phase) + PV(terminal value at end of phase 1)
Terminal value in DDM	$TV = D_{n+1} / (r-g_{long\text{-}term})$ then discount back to t=0

KEY TAKEAWAYS

Session 8 — Equity Investments

- Gordon Growth (CFA LM8): D_0=€5, g=4%, r=8% → $V=D_1/(r-g)$=5.20/0.04=€130. Common error: using D_0 not D_1.
- Union Electric preferred (CFA LM8): $4.75/0.075=$63.33. Callable issues worth less (call disadvantages investor).
- Sustainable growth: g=b×ROE. b=1-payout. Higher ROE or less payout → faster growth → higher intrinsic value.
- Justified P/E = p/(r-g). Rises with payout ratio and growth rate. Falls with higher required return.
- Canon P/E (CFA LM8): 2013-2017 range 16.6-23.9x. Peak in 2016 due to EPS collapse. 2017 near fair value at 18.8x.
- Telefónica vs DT (CFA LM8): Telefónica cheaper on P/CF (2.9x vs 3.7x) and P/E (14x vs 18x). DT cheaper on P/B.
- Porter's 5 Forces: new entrants, supplier power, buyer